

Optimal Leverage in High Frequency Trading Data

or how I learned to stop worrying and love the time average

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Overview

- 1 Motivation
- 2 Dynamic setup
- 3 First results
- 4 Stochastic Market Efficiency
- 5 More results
- 6 Final remarks

Objectives and motivation

- Extend the predictions of the Ergodicity Economics Project to shorter time scales. Why is this important?

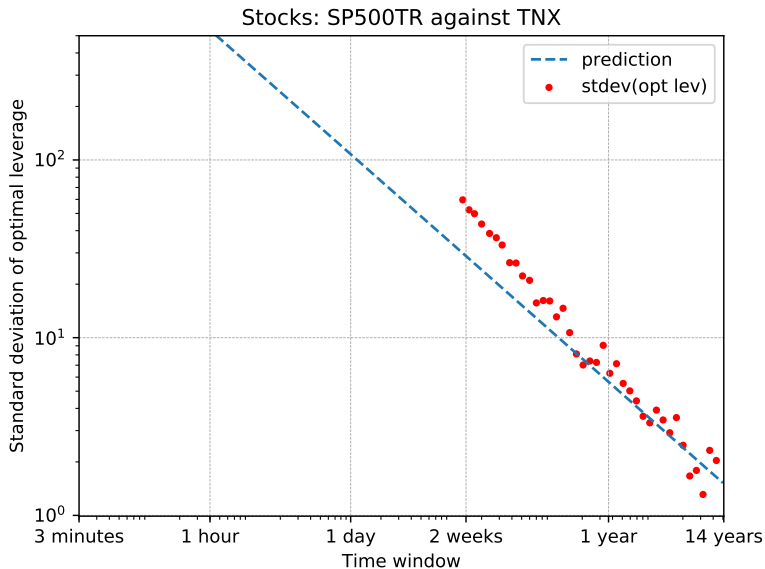
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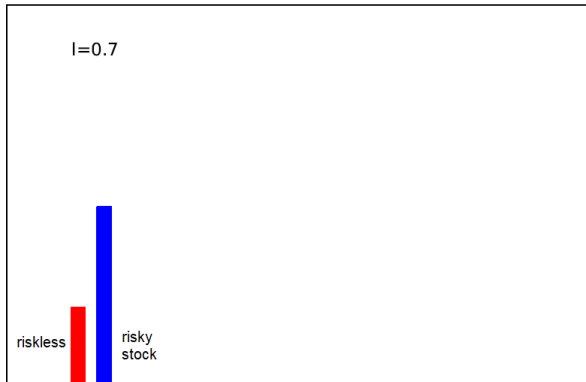
- Extend the predictions of the Ergodicity Economics Project to shorter time scales. Why is this important?
- The classic approach to market randomness involves ad hoc effects of economic information, which can hardly apply over short time scales.
- This impacts economic theory as it implicates a brand new theory of noise, solves long-standing open problems such as the equity premium puzzle, and provides other novel applications.

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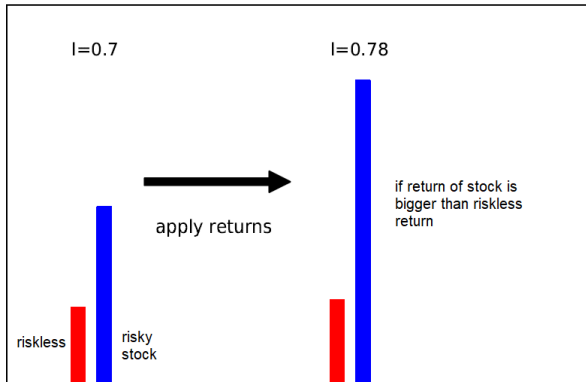
Dynamic setup

- Set up an investment on two assets: one risky stock (s) and the other riskless (r), with volatility zero. The leverage (ℓ) is the fraction of your wealth (x_ℓ) invested in the risky stock.



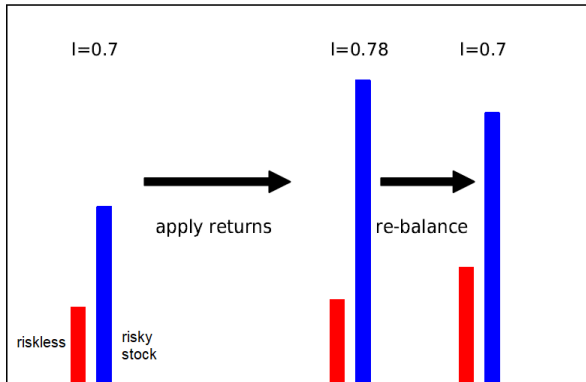
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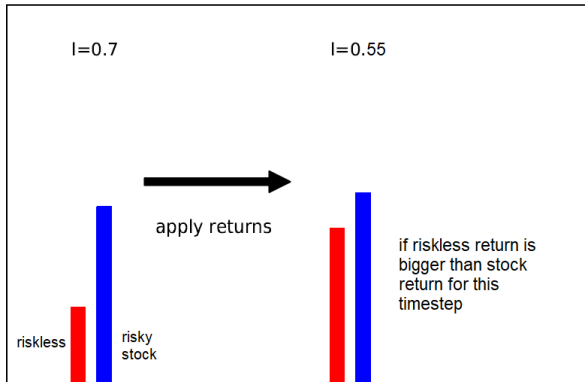
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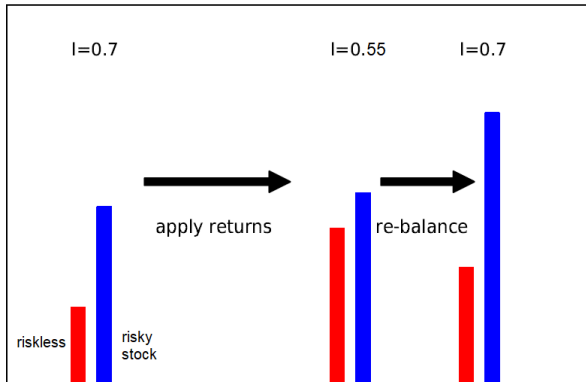
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Geometric Brownian Motion, aka GBM

Our assets will follow GBMs, a multiplicative process

$$\frac{dx}{x} = \mu dt + \sigma dW,$$

where $W \sim \mathcal{N}(0, dt)$, μ is the drift and σ the volatility.

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The growth rate of the expected wealth is

$$g_{\langle} = \frac{d \ln \langle x \rangle}{dt} = \mu$$

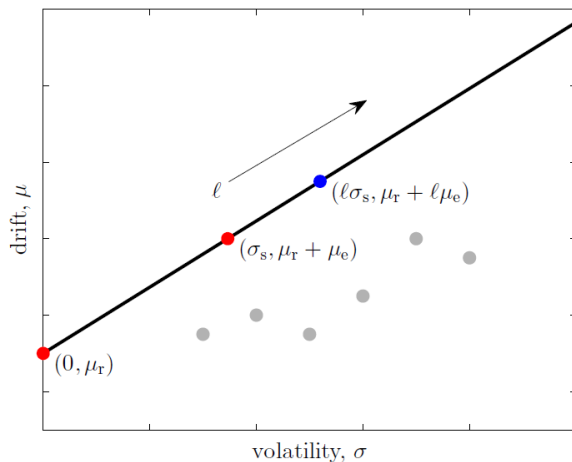
The time average growth rate is

$$\bar{g} = \lim_{\Delta t \rightarrow \infty} \frac{\Delta \ln x}{\Delta t} = \frac{d \langle \ln x \rangle}{dt} = \mu - \frac{\sigma^2}{2}$$

Wealth evolution

$$\frac{dx_\ell}{x_\ell} = (\mu_r + \ell\mu_e)dt + \ell\sigma_s dW$$

$$\mu_e = \mu_s - \mu_r$$



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- Time average growth rate approach:

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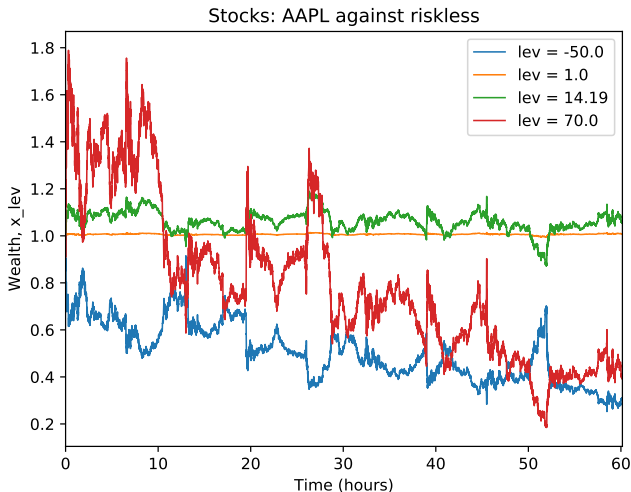
$$\ell_{\text{opt}} = \arg \max_{\ell} \bar{g} = \frac{\mu_e}{\sigma_s^2}$$

Datasets and wealth evolution

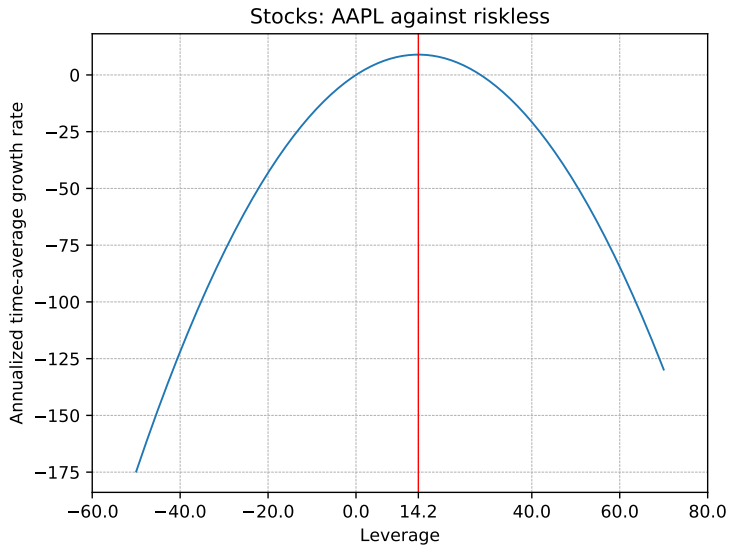
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Time average growth rate, optimal leverage



Stochastic market efficiency

Hypothesis:

- Stability argument: On sufficiently long time scales, the range $0 \leq \ell_{\text{opt}} \leq 1$ is an attractor for pairs of risky assets.
- Stochastic market efficiency: $\ell_{\text{opt}} = 1$ is an attractive point for market stochastic properties.

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Applications:

- A theory of noise

$$\sigma_s^2 = \mu_e$$

- Equity premium puzzle

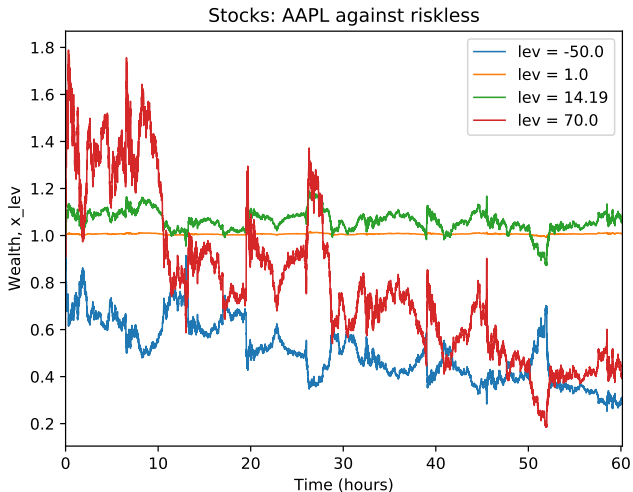
$$\pi^m = \bar{g}(1) - \bar{g}(0) = \frac{\sigma_s^2}{2}$$

Optimal leverage fluctuations

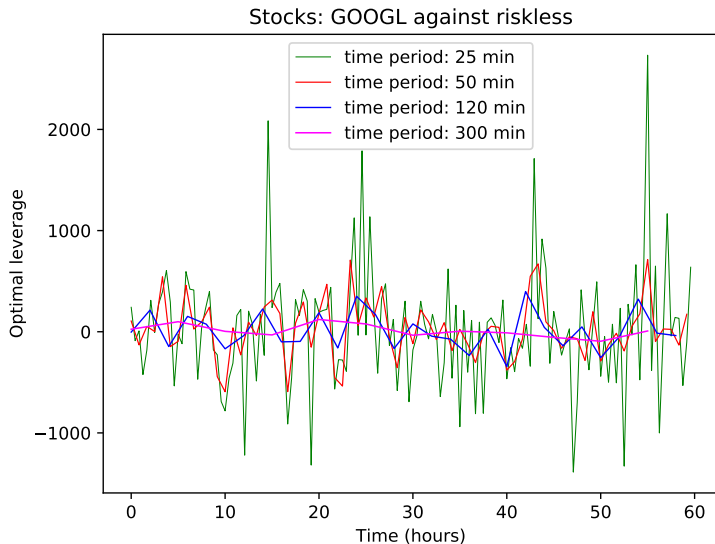
$$\text{stdev}(\ell_{\text{opt}}^{\text{est}}(\Delta t)) = \frac{1}{\sigma_s \Delta t^{1/2}}$$

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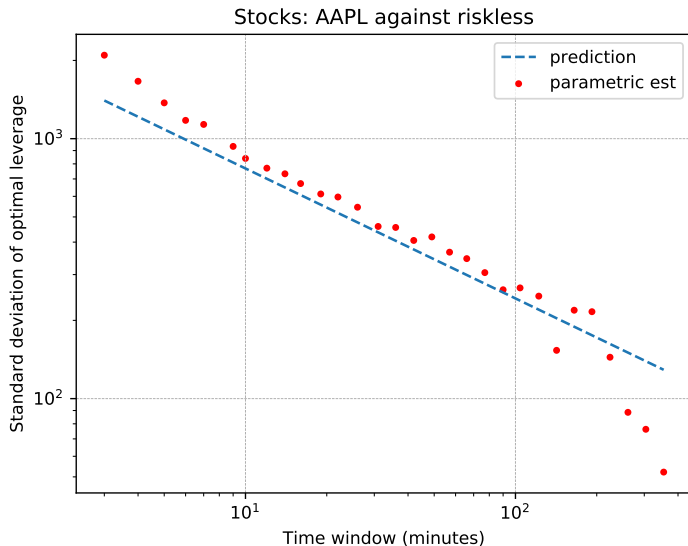
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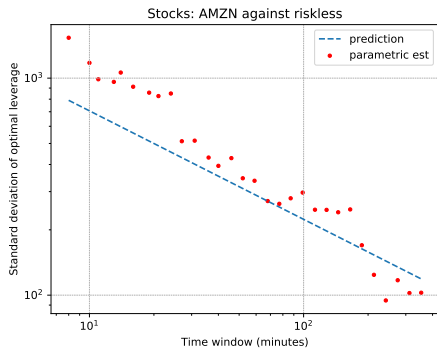
Optimal leverage time series



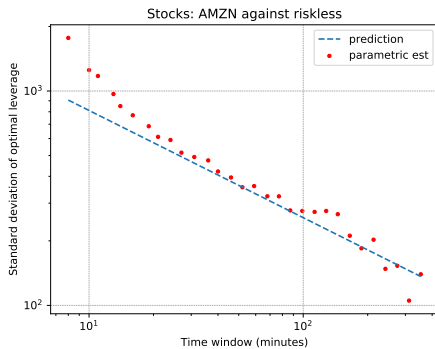
Standard deviation of optimal leverage



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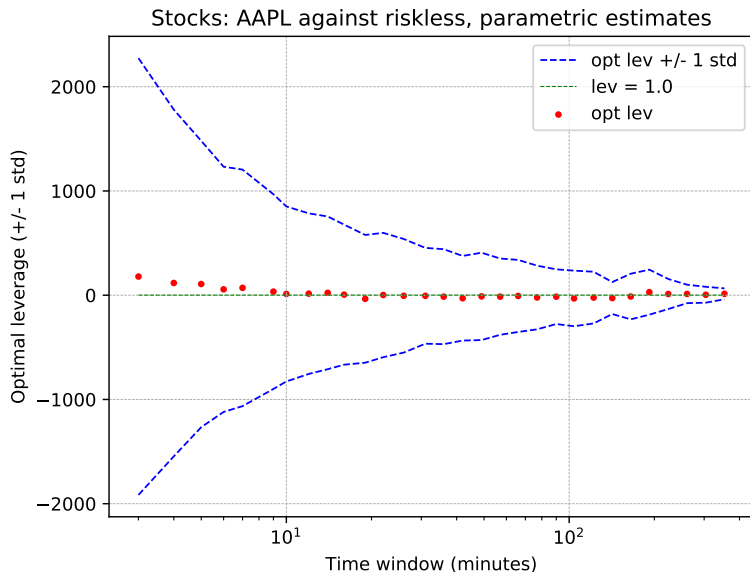


Full market days

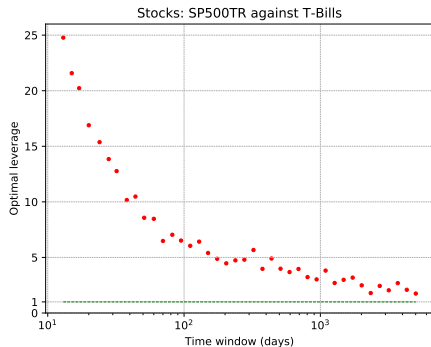
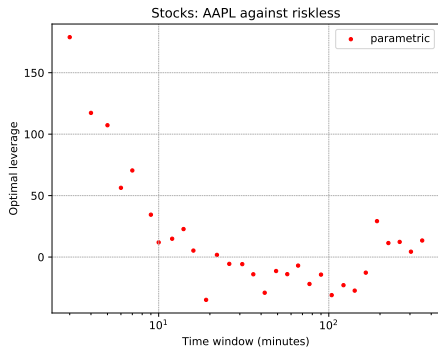


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Mean and st deviation of optimal leverage



Mean optimal leverage



- Optimal leverage and fluctuations predicted by the Ergodicity Economics Project hold remarkably well for short time scales.
- The average optimal leverage per time window is also stable and presents structure, as predicted by the stochastic market efficiency argument.
- This also supports the new theory of noise and rejects the classical arguments of economic information impact on volatility.
- Leverage efficiency is a fundamental organizing principle for the stochastic properties of markets. The data analysis in this study is an empirical test of this fundamental principle.

- O. Peters. "Optimal leverage from non-ergodicity." *Quantitative Finance*, 11(11):15931602, 2011.
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- O. Peters and A. Adamou. "Ergodicity economics lecture notes." June 2018.
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